

Comparison of ML/AD Frameworks					
Feature/Capability	DuckDB (Our Work)	PyTorch	TensorFlow	Jax	Enzyme(LLVM AD)
Automatic Differentiation	Yes	Yes	Yes	Yes	Yes
Mode Supported	Forward & Reverse	Reverse	Reverse	Forward & Reverse	Forward & Reverse
Execution Paradigm	Vectorized (Inside DBMS)	Eager/Graph	Graph	XLA Compiled	Compiler Level
Language/Interface	SQL(lambda functions)	Python	Python	Python	C/C++/LLVM IR
Data Residency	In - Database	In - Memory (External)	In - Memory (External)	In - Memory (External)	In-Process
Gradient Computation Location	Inside DuckDB Engine	External Framework	External Framework	External Framework	Compiler/Runtime
Query Optimization Integration	Yes	No	No	No	No
Scalability with Data Science(Inside DBMS)	High	Limited	Limited	Limited	Limited
Use Case Focus	Analytics+ML Inside DB	Deep Learning	Deep Learning	Research/HPC	Compiler/AD